

DRMD Best Practice (v1.0)

16.1 Producer Checklist

A. Document Identity & Administration

- ☐ Schema Versioning: The root element contains the schemaVersion attribute.
- ☐ Document Title: titleOfTheDocument is set correctly (e.g. referenceMaterialCertificate).
- ☐ Unique Identifier: A mandatory UUID is generated for the document.
- ☐ Document Identifiers: At least one external identifier (DOI, Batch ID) is included.
- ☐ Validity Period: Exactly one method is chosen to define the certificate's lifespan.
- ☐ Producer Contact: Mandatory name, email/phone, and structured location are provided.
- ☐ Responsible Persons: A list is included with at least one person flagged as mainSigner.

B. Physical Material Definition

- ☐ Material Identity: Every material entry has a human-readable name and at least one canonical materialIdentifier.
- ☐ Sample Constraints: The mandatory minimumSampleSize is defined with a numeric value and D-SI unit.
- ☐ Classification: The material is tied to a formal scheme using reference and classID.
- ☐ Multi-Material Integrity: If multiple materials exist, each is assigned a unique @id.

C. Technical Property Data (The Certified Value Core)

- ☐ Certification Flag: Every materialProperties block explicitly declares @isCertified.
- ☐ Certified Presence: At least one block has @isCertified='true' (for Profiles B/C).
- ☐ Property Identifiers: Every quantity row includes a propertyIdentifier (e.g., CAS #).
- ☐ D-SI Unit Quality: Unit strings use backslash identifiers (e.g., \gram) and meet Gold/Platinum class.
- ☐ Unit Prefix Policy: For certified values, prefixes are avoided where feasible.
- ☐ Uncertainty Completeness: Every certified value includes valueExpandedMU plus coverageFactor/Probability.
- ☐ Numeric Validation: valueExpandedMU >= 0, coverageFactor > 0, coverageProbability in (0, 1].
- ☐ Consistency: A single result table does not mix list-encoded and row-encoded quantities.
- ☐ Traceability: A structured measurementMetaData block or narrative statement confirms traceability to SI.

D. Statements & Archival

- ☐ Mandatory Narrative: intendedUse, storageInformation, and instructionsForHandlingAndUse are non-empty.
- ☐ Safety Info: Hazards are documented in healthAndSafetyInformation.
- ☐

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Embedded PDF: Profile C documents include a Base64-encoded PDF.

☐ Digital Signature: At least one ds:Signature is applied.